POLITECNICO
MILANO 1863

BASI DI DATI 1
Anno Accademico 2023-2024
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Appello 24/02/2024 **DURATION: 2 ore**

Part 1 – QUERY LANGUAGES

DronEat is a startup that delivers food to Polimi students anywhere on campus using drones.

INGREDIENT (IdIngr, Name, KCalPerGram)
DISH (IdDish, Name, Price)
RECIPE (IdDish, IdIngr, QuantityInGrams)
ORDER (IdOrder, IdDish, IdStudent, OrderDate, OrderTime, DeliveryCoordinates, ActualDeliveryTime)
STUDENT (IdStudent, Lastname, Firstname, City)

Exercise 1 – SQL (13pt)

A. Define the schema of ORDER, including referential integrity constraints and attributes nullability. Use the POINT data type for coordinates. (2pt)
B. Find the name and ID of the dish (or dishes) with the highest total number of calories. (4 pt)
C. Find the names of the students who, considering all their orders collectively, have ordered at most two different types of dishes containing only ingredients with a calorie count exceeding 2 grams. (4 pt)
D. Find, for each dish, the date and time of its first order. (Note: IdOrder does not have values generated in chronological order) (4 pt)

Exercise 2 - Formal Languages (6pt)

Express the following query in optimized relational algebra and Datalog:
E. Find the names of students who have ordered only dishes that contain both pepper and onion.

PART 2 – DATABASE DESIGN

Exercise 3 - Theory (1pt)

Explain the concept of unique key in a relational database. What is its purpose and how does it differ from the primary key?

Exercise 4 - Conceptual and Logical Design (13pt)

• The project is evaluated for completeness, correctness, readability, conciseness, and self-explanatoriness.
• Remember to specify the identifiers of all entities and the cardinalities of all associations

A regional tourist agency manages all the information to enable online search and booking of summer residences (houses and villas). Each residence is characterized by its location, the number of rooms and beds, the distance from the sea, and whether there is a sea view. Some residences have their own website (indicated by URL); others have a set of photographs and descriptive texts managed by the agency and presented online. Rentals are established on a weekly basis; owners designate the weeks available for rent, and each week has a different price. Residences also offer a range of additional services (e.g., extra cleaning, airport transfers, room service breakfast) that customers can select during the booking process. Each extra service has a variable cost depending on the residence and can be selected by the customer along with the residence booking.
Customers are characterized by name, address, telephone number, and email. The booking process involves selecting the desired residence for a specific week, which is secured by paying a 30% deposit; the remaining 70% balance must be settled within 4 weeks of entering the residence. Once the residence is booked, the corresponding week is no longer visible on the agency's website. At the beginning of the stay, an additional amount is paid to cover any damages, which is refunded at the end of the stay; the amount is associated with each residence. In case of damages, the withheld amount and a description of the damages caused are recorded.